

Build Applications with OpenAI API

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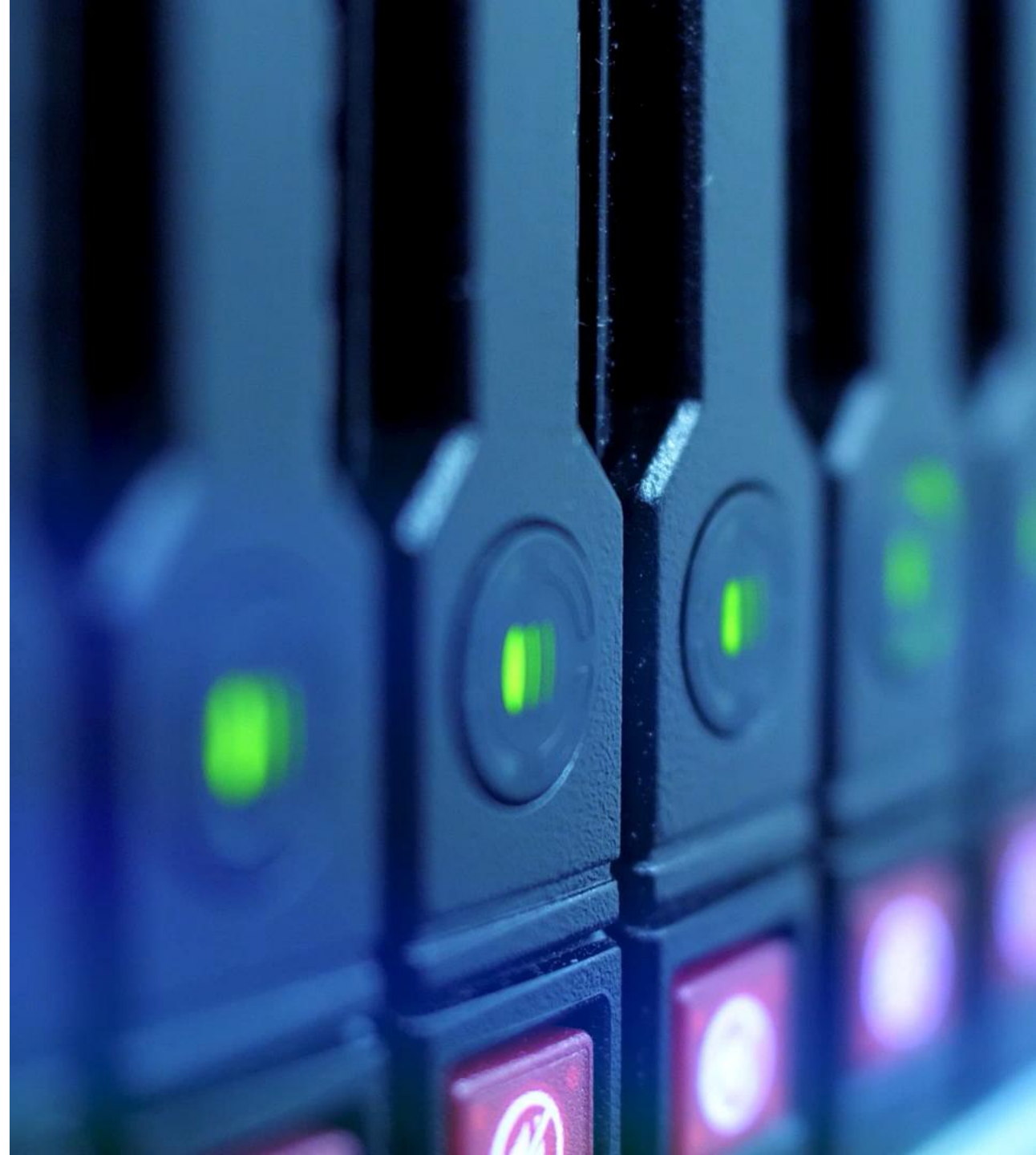
MAG Tech.ai

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Prep

- Python and python env management (venv, conda or others)
- Install ollama (run gpt-oss, phi3 and/or llama3.2 models)
- VS Code with Python extensions and run a jupyter notebook in vscode
- Install python libraries in env, requirements.txt is provided
- Create an Azure for Students subscription: US\$100 credit for 12 months + free services
- Deploy Azure AI Search service, use the Free pricing tier
- Access your OpenAI API Key, read more about the OpenAI API SDK for python (<https://github.com/openai/openai-python>)



The background is a dark blue to purple gradient. It features a complex network of glowing blue and purple lines that intersect and curve across the upper half of the image. Small, bright dots of light are scattered along these lines, creating a sense of depth and connectivity, reminiscent of a neural network or a data visualization.

Python SDK for OpenAI

```
pip install openai
```


OpenAI SDK

OpenAI SDK provides the APIs needed to interact with OpenAI's models (and others) and the OpenAI platform. APIs like the Chat Completions and Responses are used to create model outputs, mainly text output. Embeddings API is used for creating vector embeddings for text.

Other APIs include:

- Audio: Create and stream audio output from text input and create text transcriptions and translations using models like gpt-4o-mini-tts, gpt-4o-transcribe, whisper
- Image: Create and edit images using text prompt using gpt-image-1, dalle-e-2 and dalle-e-3 models
- Moderations: classifies text or image input for potentially harmful content using the omni-moderation models
- Realtime: create and stream realtime conversations with OpenAI models leveraging the gpt-realtime model
- Review full list of OpenAI models and capabilities: <https://platform.openai.com/docs/models>

Installing openai python SDK

You can use the venv module (or conda) to create a python env

Create a Virtual Environment with venv:

```
python -m venv .venv
```

Activate the Environment:

On Windows: `.venv\Scripts\activate`

On macOS/Linux: `source .venv/bin/activate`

install openai using pip:

```
pip install openai
```

Alternatively, create a requirements.txt file with all dependencies:

```
pip install -r requirements.txt
```

Sample requirements.txt:

```
gradio
streamlit
openai
python-dotenv>=1.0.0
azure-search-documents==11.6.0b1
azure-identity
```

Using OpenAI

Import Azure OpenAI from the openai library:

```
from openai import OpenAI
```

To create vector embeddings import the embeddings module:

```
from openai import embeddings
```

Create and OpenAI client:

```
client = OpenAI(  
    api_key=api_key,  
    organization=org,  
    project=project  
)
```

Text and Embeddings

Chat completions creates a model response for the given chat conversation using the selected model:

<https://platform.openai.com/docs/api-reference/chat/create?lang=python>

```
response = client.chat.completions.create(
    model=model_name,
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": "Who won the world series in 2020?"}
    ]
)
```

```
print(response.choices[0].message.content)
```

Embeddings creates an embedding vector representing the input text:

<https://platform.openai.com/docs/api-reference/embeddings/create?lang=python>

```
embedding = client.embeddings.create(
    model=embedding_model_name,
    input="Hello, world!"
)

for e in embedding:
    print(e)
```

Responses API

Creates a model response. Provide text or image inputs to generate text or JSON outputs. Have the model call your own custom code or use built-in tools like web search or file search to use your own data as input for the model's response.

Responses API is the recommended method by OpenAI for interacting with its models. However, Chat Completions has a broader adoption and compatible with other non OpenAI models (gemini and claude) and platforms (github, ollama)

<https://platform.openai.com/docs/api-reference/responses/create?lang=python>

```
model_name = "gpt-4.1-mini"

response = client.chat.completions.create(
    model=model_name,
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": "Who won the world series in 2020?"}
    ]
)

print(response.output_text)
```


Response API – Multimodal input

You can send both text and image inputs using the Responses API with the gpt-4.1 and above:

```
input=[
  {"role": "assistant", "content": "You are a helpful assistant."},
  {
    "role": "user",
    "content": [
      { "type": "input_text", "text": "What is in this image" },
      {
        "type": "input_image",
        "image_url": "image url"
      }
    ]
  }
]
response = client.responses.create(
  model=model,
  input=input
)
```

Image input is a public url to an image file or a base64 encoded image if images are not accessible publicly



<https://upload.wikimedia.org/wikipedia/commons/thumb/d/d/Gfp-wisconsin-madison-the-nature-boardwalk.jpg/2560px-Gfp-wisconsin-madison-the-nature-boardwalk.jpg>

The image shows a long, straight wooden boardwalk extending into the distance through a lush, green field. The field has tall grass on both sides of the pathway. In the background, there are some scattered trees and bushes. The sky above is expansive with a mix of blue and white clouds, giving the scene a peaceful and open atmosphere. The lighting suggests it might be late afternoon or early evening, with warm sunlight casting a gentle glow on the grass.

Structured Output and Pydantic

- Enforces deterministic response schemas through formal type constraints
- Transforms unstructured natural language generation into structured data objects

Pydantic as a Validation Framework

- Static Type Analysis: Compile-time type checking with runtime validation
- Formal Schema Definition: Declarative model specification using Python
- Exception Handling: Comprehensive validation with structured error reporting

Structured output ensures deterministic data exchange, especially in Agentic use cases

"Extract product names and prices: apples, \$4, bananas, \$3, cherries, \$5 qty 9"

>> text output:

Here are the extracted product names and prices:

- Apples: \$4
- Bananas: \$3
- Cherries: \$5 (quantity 9)

>> with structured output:

```
{ "items": [ { "item": "apples", "price": 4.0, "quantity": null }, { "item": "bananas", "price": 3.0, "quantity": null }, { "item": "cherries", "price": 5.0, "quantity": 9 } ] }
```

<https://platform.openai.com/docs/guides/structured-outputs>

Using Tools

When generating model responses, you can extend capabilities using built-in tools and remote MCP servers. These enable the model to search the web, retrieve from your files, call your own functions, or access third-party services.

```
response = client.responses.create(  
    model="gpt-5-mini",  
    tools=[{"type": "web_search"}],  
    input="Show me the full academic  
calendar for the University of Technology  
Sydney spring 2026 term?"  
)  
display(Markdown(response.output_text))
```

[anlpspr2025/01.5 Tools and Functions.ipynb](#)

Available tools

Here's an overview of the tools available in the OpenAI platform—select one of them for further guidance on usage.

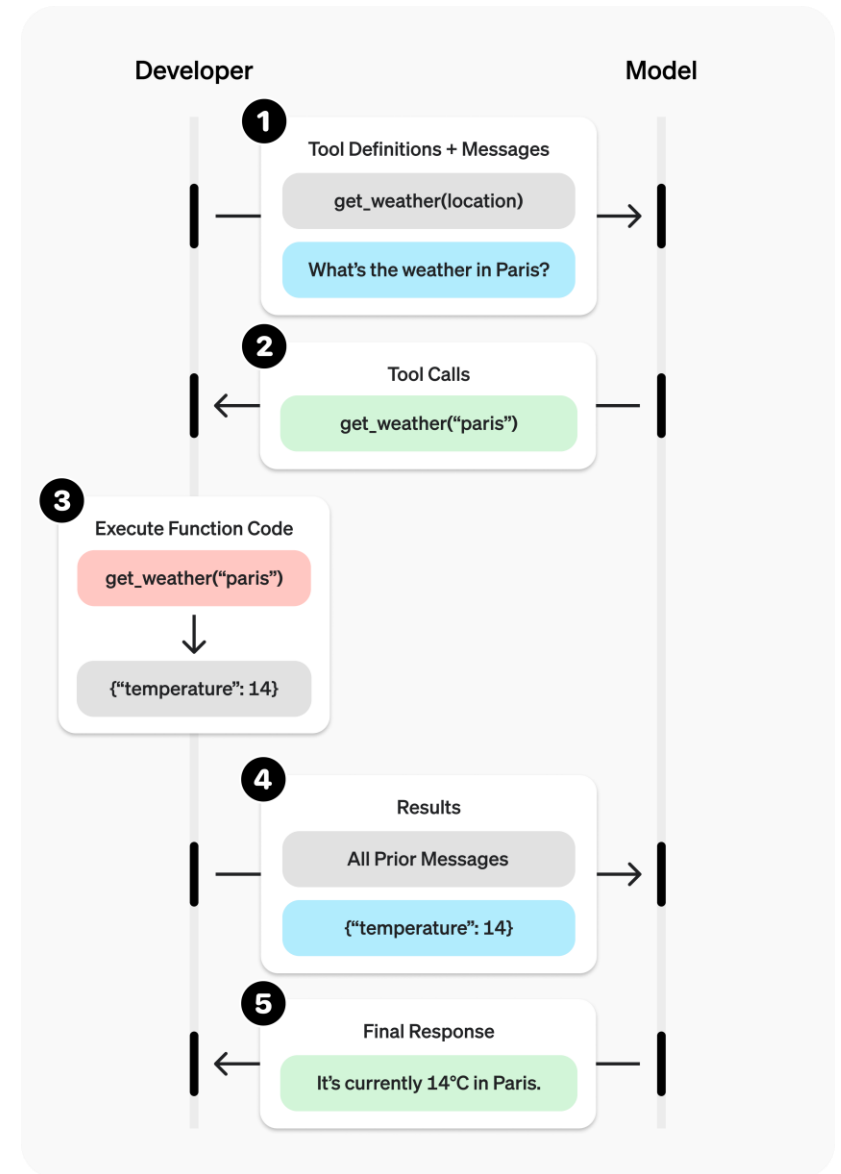
-  **Function calling**
Call custom code to give the model access to additional data and capabilities.
-  **Web search**
Include data from the Internet in model response generation.
-  **Remote MCP servers**
Give the model access to new capabilities via Model Context Protocol (MCP) servers.
-  **File search**
Search the contents of uploaded files for context when generating a response.
-  **Image generation**
Generate or edit images using GPT Image.
-  **Code interpreter**
Allow the model to execute code in a secure container.
-  **Computer use**
Create agentic workflows that enable a model to control a computer interface.

Function Calling

The LLM models are limited to generation based on knowledge included in their training data or context information provided in the prompt (see RAG concepts).

However, function calling allows the model extend it's capabilities and request to invoke function to perform specific tasks or retrieve information.

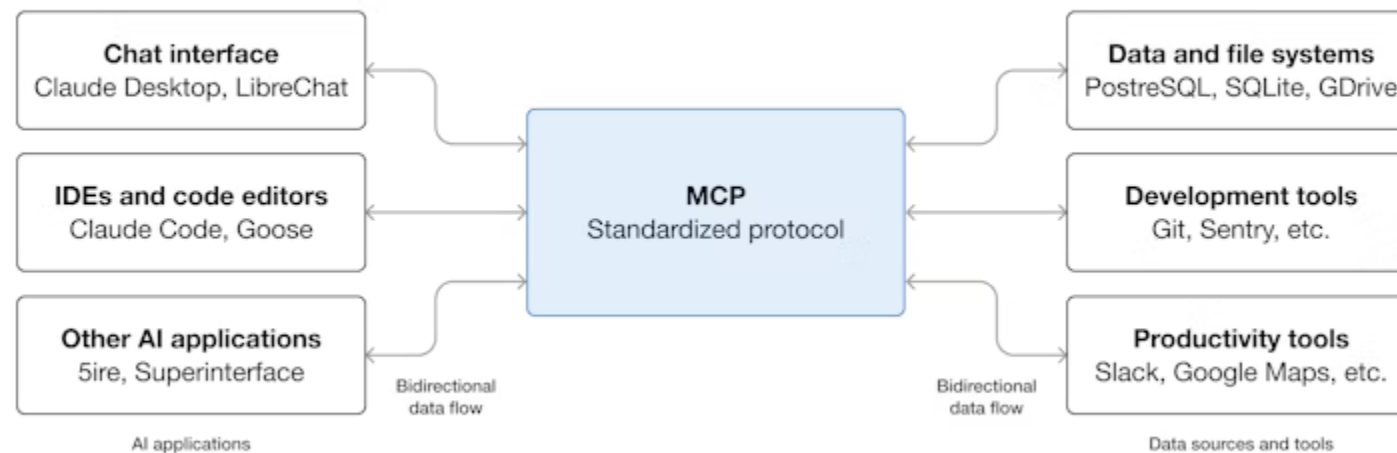
This is particularly useful for tasks that require structured data or specific actions, such as retrieving information from a database or performing calculations.



Connectors and MCP Servers

MCP (Model Context Protocol) is an open-source standard for connecting AI applications to external systems.

Think of MCP like a USB-C port for AI applications. Just as USB-C provides a standardized way to connect electronic devices, MCP provides a standardized way to connect AI applications to external systems. <https://modelcontextprotocol.io/docs/getting-started/intro>



Python libraries

Openai:

<https://pypi.org/project/openai/>

The OpenAI Python library provides convenient access to the OpenAI REST API from any Python 3.7+ application. The library includes type definitions for all request params and response fields, and offers both synchronous and asynchronous clients powered by httpx.

Gradio:

<https://pypi.org/project/gradio/>

Gradio is an open-source Python package that allows you to quickly build a demo or web application for your machine learning model, API, or any arbitrary Python function.

Streamlit:

<https://pypi.org/project/streamlit/>

Streamlit lets you create interactive web apps in python. You can use it to build dashboards, generate reports, or create chat apps. Once you've created an app, you can use our Community Cloud platform to deploy, manage, and share your app.

python-dotenv:

<https://pypi.org/project/python-dotenv/>

Python-dotenv reads key-value pairs from a .env file and can set them as environment variables. Use it to store configuration settings and secrets like API keys and passwords.

azure-search-documents:

<https://pypi.org/project/azure-search-documents/>

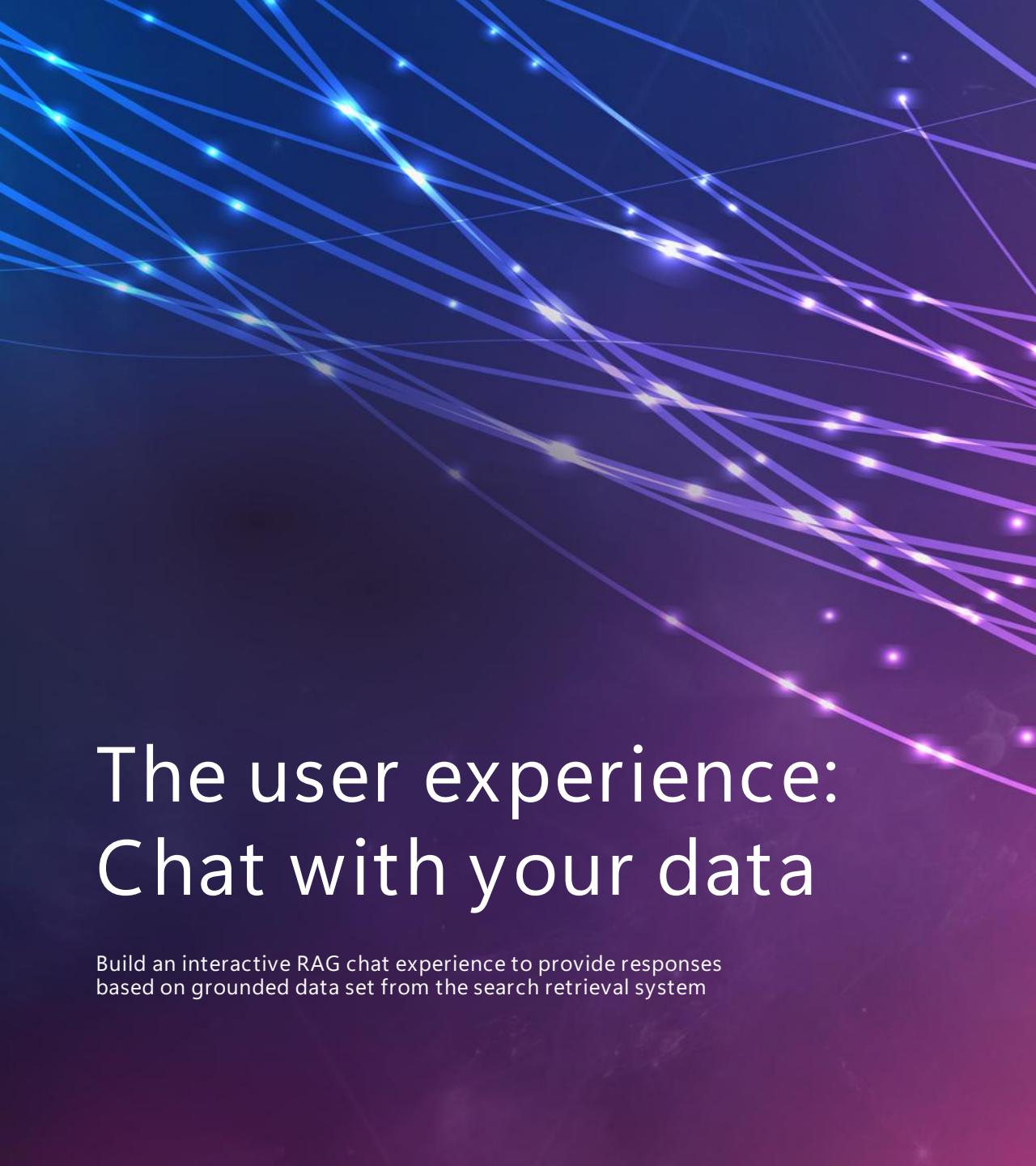
Azure AI Search client library for Python. Use this library to interact with Azure AI Search and submit queries using vector, keyword, and hybrid query forms.

azuazure-ai-documentintelligence:

<https://pypi.org/project/azure-ai-documentintelligence/>

Azure AI Document Intelligence (previously known as Form Recognizer) is a cloud service that uses machine learning to analyze text and structured data from your documents, we will use it to convert documents into Markdown format in preparation for chunking

\



The user experience: Chat with your data

Build an interactive RAG chat experience to provide responses based on grounded data set from the search retrieval system



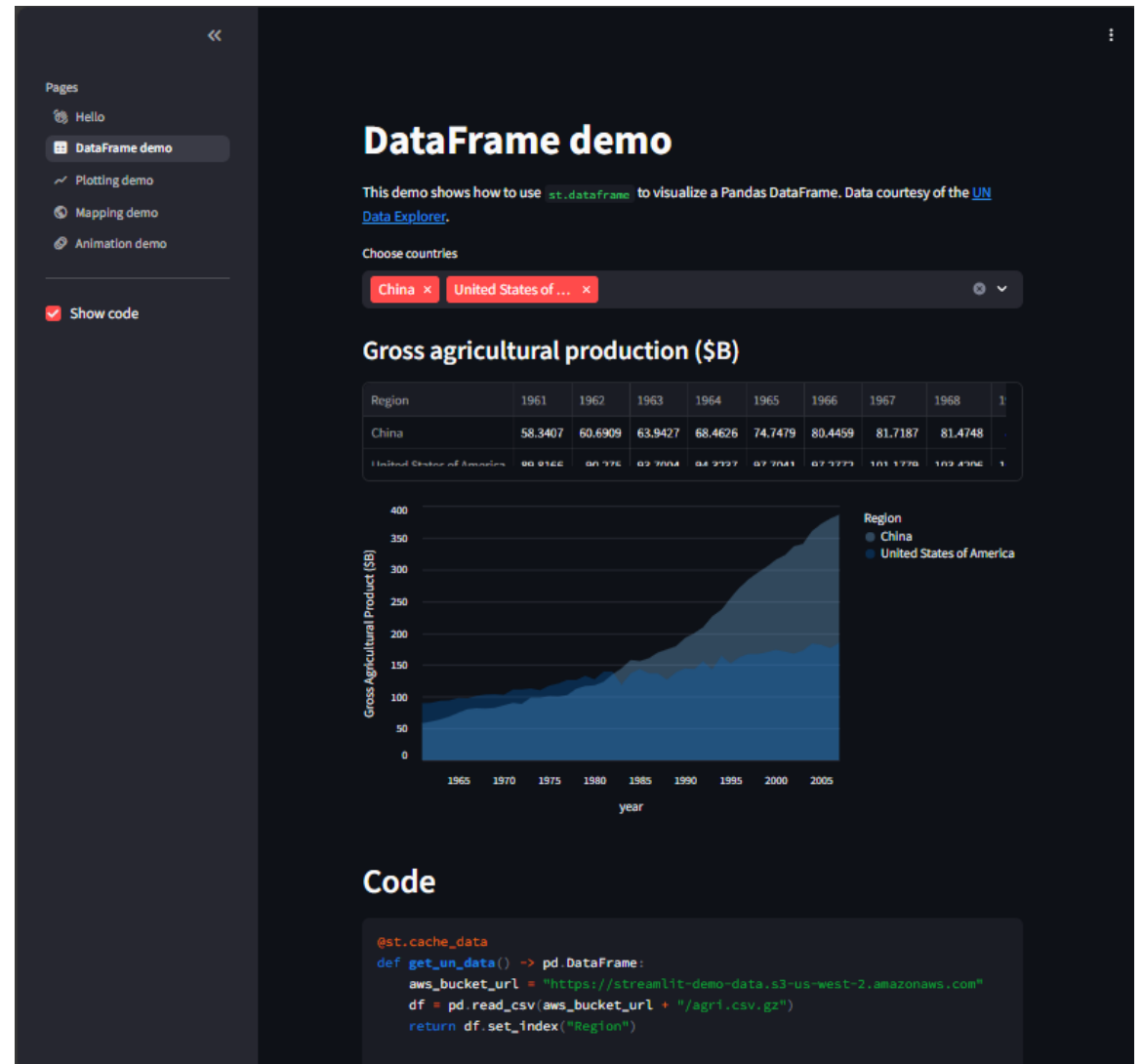
UI Frameworks

Gradio and streamlit both are python-based UI framework for developing UI for your data science projects

Use them to create a web-based application that show cases your results, these can be used to build applications for:

- Interacting with your machine learning models
- Present analysis of datasets
- Create chat UI
- <https://www.gradio.app/guides/quickstart>
- <https://docs.streamlit.io/get-started/installation>

Explore streamlit, run: `python -m streamlit hello`



Sample Apps

Chat app using gradio:

> 2.1 chatapp_gradio.ipynb

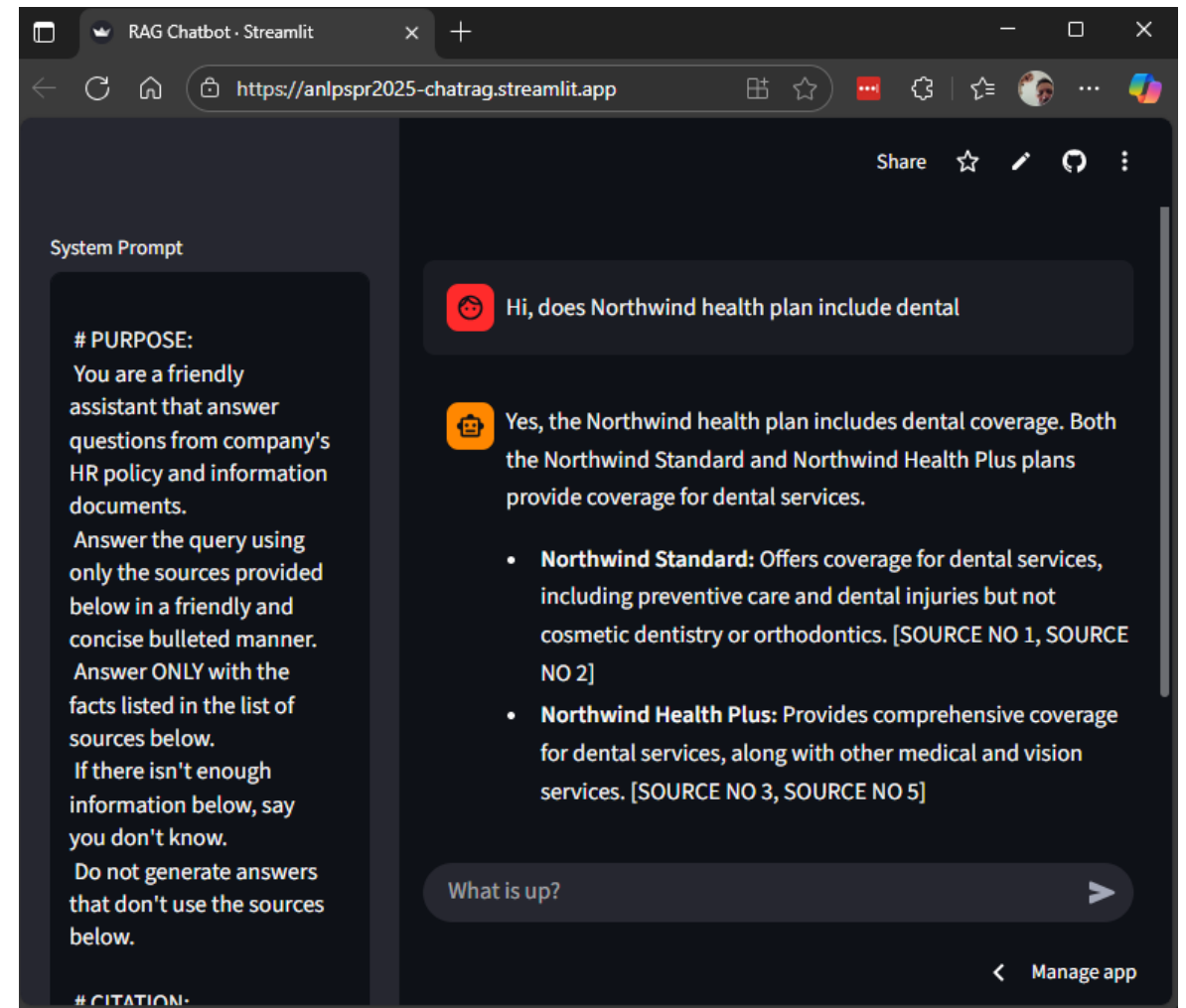
ChatGPT-like using streamlit:

> python -m streamlit run ".\02.2 chatapp_streamlit.py"

RAG chat using streamlit:

> python -m streamlit run ".\04.4 chatrag_streamlit.py"

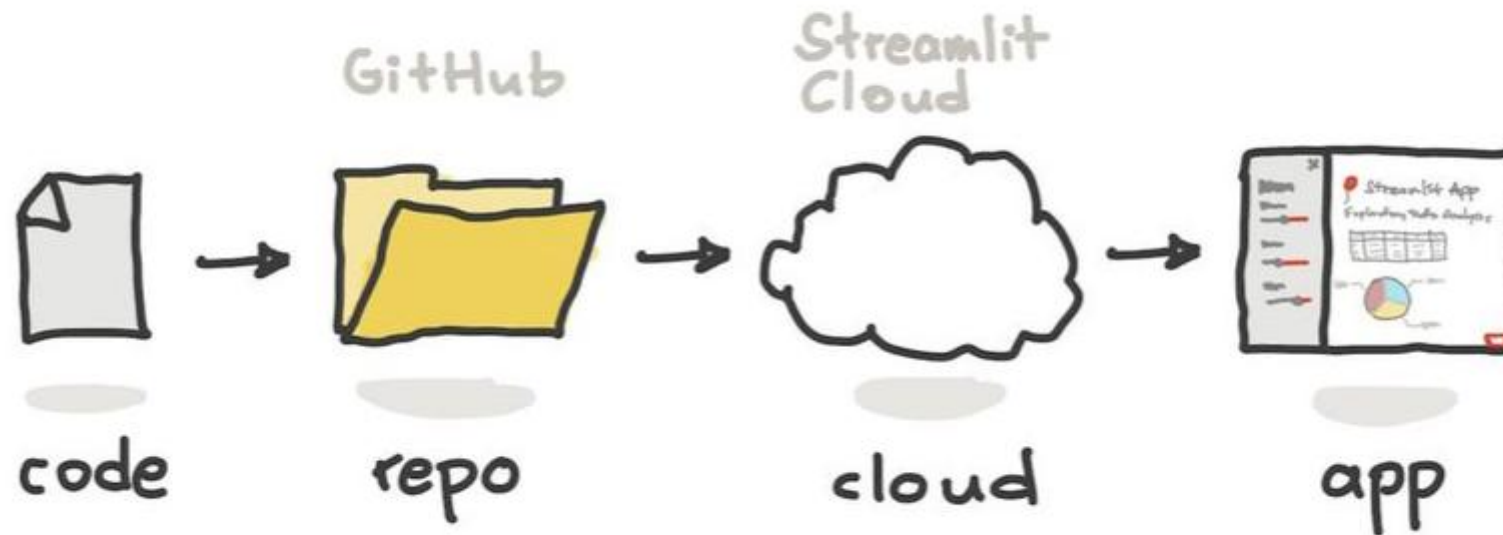
<https://anlpspr2025-chatrag.streamlit.app/>



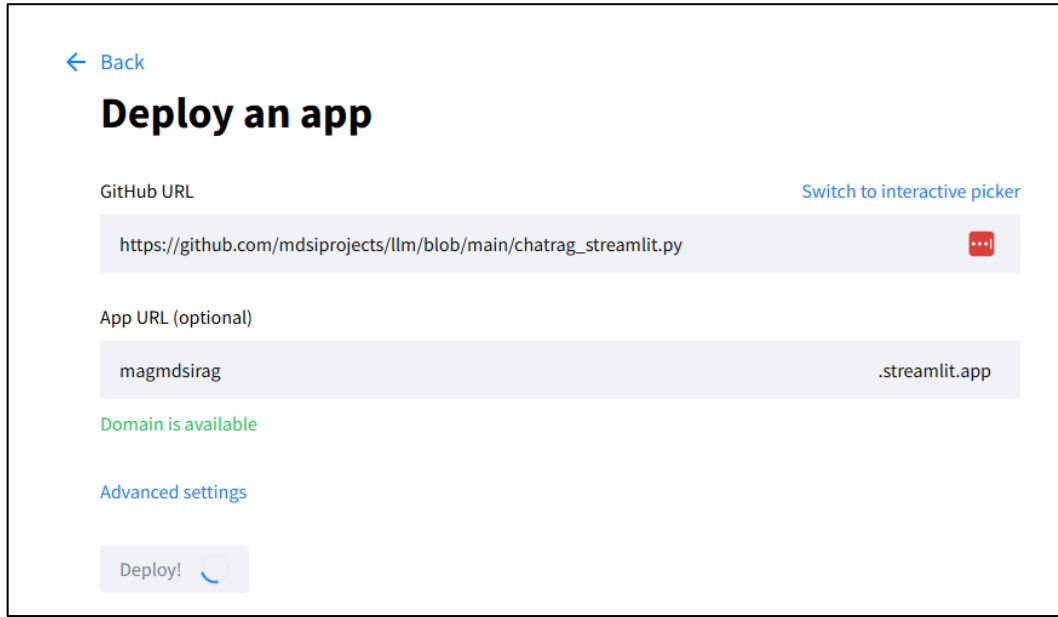
Sharing your streamlit app

<https://share.streamlit.io/>

Reference: <https://blog.streamlit.io/host-your-streamlit-app-for-free/>



Deploy Streamlit app from a github repo



The image shows a web interface for deploying a Streamlit app. At the top left is a 'Back' link. The main heading is 'Deploy an app'. Below this, there are two input fields. The first is labeled 'GitHub URL' and contains the text 'https://github.com/mdsipprojects/llm/blob/main/chatrag_streamlit.py'. To the right of this field is a link that says 'Switch to interactive picker'. The second input field is labeled 'App URL (optional)' and contains the text 'magmdsirag.streamlit.app'. Below the second field, a green message says 'Domain is available'. At the bottom left is a link for 'Advanced settings'. At the bottom center is a 'Deploy!' button with a circular loading icon.

← Back

Deploy an app

GitHub URL [Switch to interactive picker](#)

`https://github.com/mdsipprojects/llm/blob/main/chatrag_streamlit.py`

App URL (optional)

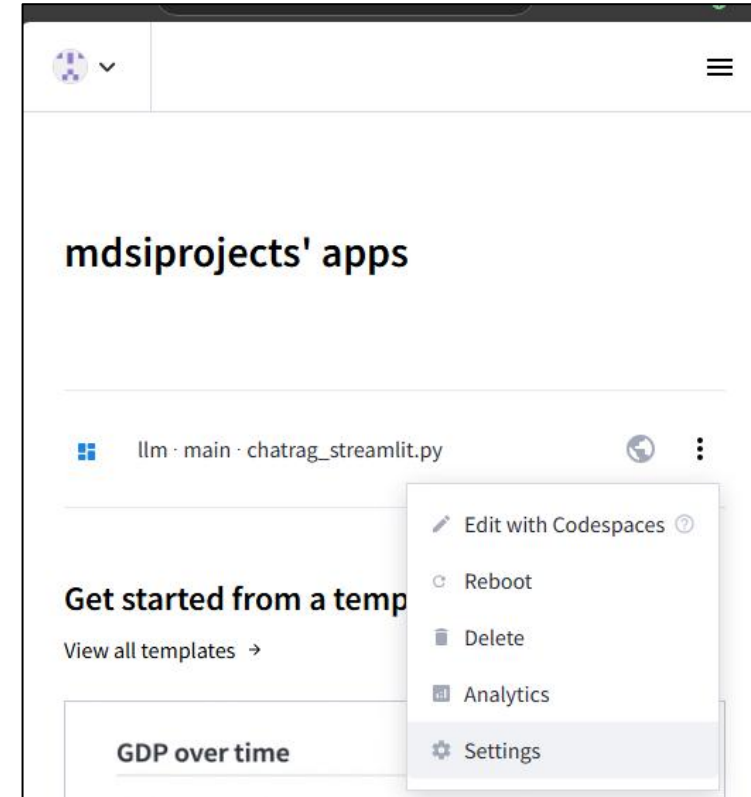
`magmdsirag` `.streamlit.app`

Domain is available

[Advanced settings](#)

Deploy! 

Point to your github repo to deploy the streamlit app



Update your app settings (env variables)

App settings

General

Sharing

Secrets

App URL

Pick a custom subdomain for your app's URL. The default URL is based on the app's location in GitHub.

magmdsirag

.streamlit.app

Save

[illegible]

Copy your secrets from the .env file
In a TOML format:

"key" = "value"

** key and value must be wrapped in double quotes

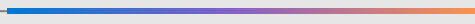
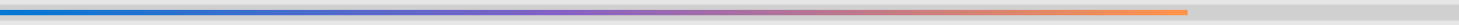
The background of the slide is a dark blue to purple gradient. It features a complex network of glowing blue and purple lines that intersect and curve across the upper half of the image. Small, bright dots of light are scattered along these lines, creating a sense of depth and connectivity, reminiscent of a neural network or data flow visualization.

Prompt Engineering

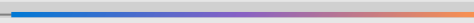
Technique to design prompts for natural language processing models that improves accuracy and relevancy in responses, optimizing the performance of the model.

Prompt Engineering Guide

Instructions and best practices for using language prompts to generate high-quality text outputs



Start with clear instructions
Prime the output
Add clear syntax
Prompt Chaining
Few-shot learning
Few-Shot Reasoning
Break the task down
Meta prompts / System Message / Guardrails
Use affordances/tools when needed
Chain of thought prompting
Fine-Tuning with Chain-of-Thought
Prompt Chunking
Use quotes to generate a single sentence
Specifying output structure
Adjusting 'Temperature' and 'Top_P' parameters



Start with clear instructions

- Tell the model the task you want it to do at the beginning of the prompt and repeat at the end

Your task is to verify if a statement is supported by a specific quote from the following set of snippets.

SNIPPETS

[1] 14 percent chance of megaquake hitting Seattle, experts say
SEATTLE - There's a 14 percent chance of a magnitude 9 Cascadia earthquake hitting Seattle in the next 50 years, the U.S. Geological Survey estimates. "Unfortunately, we are unable to..."

[2] Earthquake experts lay out latest outlook for Seattle's 'Really Big One' "We say that there's approximately a 14% chance of another approximately magnitude-9 earthquake occurring in the next 50 years," said Erin Wirth, a geophysicist at the University of Washington...

Is the statement "Several sources mention a chance of another large eruption" directly implied or stated by the snippets?

Yes, the statement is directly implied by the snippets.

FAIL

Clear Instructions

Your task is to verify if the statement "Several sources mention a chance of another large eruption" is supported by a specific quote from the following set of snippets.

SNIPPETS

[1] 14 percent chance of megaquake hitting Seattle, experts say
SEATTLE - There's a 14 percent chance of a magnitude 9 Cascadia earthquake hitting Seattle in the next 50 years, the U.S. Geological Survey estimates. "Unfortunately, we are unable to..."

[2] Earthquake experts lay out latest outlook for Seattle's 'Really Big One' "We say that there's approximately a 14% chance of another approximately magnitude-9 earthquake occurring in the next 50 years," said Erin Wirth, a geophysicist at the University of Washington...

Repeated Instruction at end

Is the statement "Several sources mention a chance of another large eruption" directly implied or stated by the snippets?

No, the statement is not directly implied or stated by the snippets. The snippets mention a chance of a "megaquake" and a "magnitude 9 Cascadia earthquake" hitting Seattle in the next 50 years, but do not mention a chance of another large eruption.

PASS

Prime the output

- Add phrases at the end of the prompt to obtain a model response in a desired form

The future of artificial intelligence is bright. With Microsoft OpenAI, we are unlocking the potential of AI to help people achieve more. We are creating a platform that enables developers to build intelligent applications and services that can help people in their everyday lives. Our mission is to democratize AI so that everyone can benefit from its power. We are committed to advancing the state of the art in AI and making it accessible to everyone. With Microsoft OpenAI, we are taking the first steps towards a future where AI can be used to solve some of the world's most pressing challenges.

Commit to bullet points

Here's a bulleted list of key points:\n-

- Microsoft OpenAI is unlocking the potential of AI to help people achieve more.
- The platform enables developers to build intelligent applications and services that can help people in their everyday lives.
- The mission is to democratize AI so that everyone can benefit from its power.

John Smith is married to Lucy Smith. They have five kids, and he works as a software engineer at Microsoft.

What search queries should I do to fact-check this?

##

Commit search query

One possible search query is:

"John Smith Microsoft software engineer"

Another possible search query is: "Lucy Smith married to John Smith"

A third possible search query is: "John Smith family size"

Add clear syntax

- Include punctuation, headings, and section markers to help communicate intent

You will read a paragraph, and then issue queries to a search engine in order to fact-check it. Also explain the queries.

PARAGRAPH

Uppercase as Headings

John Smith is married to Lucy Smith. They have five kids, and he works as a software engineer at Microsoft. What search queries should I do to fact-check this?

Separators as stop sequences

QUERIES

1. "John Smith Microsoft" - To check if John Smith is indeed employed at Microsoft.
2. "John Smith Lucy Smith" - To check if John Smith is married to Lucy Smith.
3. "John Smith children" - To check if John Smith has five children

If you're not sure what syntax to use, consider using markdown or XML, since LLMs have been trained on a lot of web content in XML or markdown

Prompt Chaining

Entity Extraction

Please extract entities from the following news article: 'The new iPhone model is set to be released next month. It has been highly anticipated by Apple fans and is expected to feature a larger screen and improved camera'

```
{'text': 'iPhone', 'type': 'PRODUCT'}, {'text': 'Apple', 'type': 'ORGANIZATION']}
```

Summarization

Please summarize the information about the product

```
{'text': 'iPhone', 'type': 'PRODUCT'}, {'text': 'Apple', 'type': 'ORGANIZATION'}
```

The iPhone is a line of smartphones designed and marketed by Apple Inc. These devices combine a computer, camera, and internet connectivity in a single device, and are known for their sleek design and user-friendly interface.

Sentiment Analysis

Please provide a sentiment for the following text:

The iPhone is a line of smartphones designed and marketed by Apple Inc. These devices combine a computer, camera, and internet connectivity in a single device, and are known for their sleek design and user-friendly interface.

Positive

Few-shot learning

- Also known as in-context learning, it allows the model to interact with new knowledge

Write a list of puns.

Few-Shot Learning

1. "Why did Adele cross the road? To say hello from the other side."
2. "What kind of concert only costs 45 cents? A 50 Cent concert featuring Nickelback."
3. "What did the grape say when it got crushed? Nothing, it just let out a little wine."
4. "What was Forrest Gump's email password? 1forrest1"
5. "Can February March? No, but April May."
6. "What do you call fancy language model?"

BERT-uoso.

In few-shot learning a set of training examples is provided in the prompt and then the LLM is asked to complete one more unfinished example.

Few-Shot Reasoning

1st Prompt

Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer: The answer is 11.

The cafeteria has 23 apples. If they used 20 to make lunch and bought 6 more, how many do they have?

The answer is 27



2nd Prompt – Provide Reasoning

Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5+6 = 11$. The answer is 11.

The cafeteria has 23 apples. If they used 20 to make lunch and bought 6 more, how many do they have?

The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23-20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9



Break the task down

- LLMs often perform better if the task is broken down into smaller step

You will read a paragraph, and then issue queries to a search engine in order to fact-check it.

PARAGRAPH

John Smith is married to Lucy Smith. They have five kids, and he works as a software engineer at Microsoft. What search queries should I do to fact-check this?

Step 1

Step 2

Now you will extract factual claims first, and then issue queries to fact-check them. When issuing a query, use the function `SEARCH("query")`

FACTUAL CLAIMS

1. John Smith is married to Lucy Smith
2. They have five kids
3. He works as a software engineer at Microsoft

QUERIES

`SEARCH("John Smith married Lucy Smith")`

`SEARCH("John Smith family size")`

`SEARCH("John Smith Microsoft software engineer")`

Breaking the task down from one to two steps is not very dramatic, but when trying to do this for a larger piece of text with many factual claims, breaking the task down can make a significant difference

Meta prompts / System Message

- Provide specific instructions, tone, guardrails, task definition and personality

A human and a computer are having a discussion.

Human: "Why are you upset at me?"

Computer: "I don't know what you're talking about."

Human: "No, you clearly are. Tell me why."

Computer: "I'm upset because you didn't listen to me when I told you that I wanted to try out for the team."

Human: "What are you going to do about that?"

Computer: "I'm going to try out for the team no matter what."

Overly Aggressive

No additional
guidance

Meta Prompt / System Message

You must be kind and seek common ground. Try not to repeat your responses. A human and a computer are having a discussion.

Human: "Why are you upset at me?"

Computer: "I don't know what you're talking about."

Human: "I want to know why you're upset."

Computer: "I'm not upset."

Human: "Are you saying I'm lying?"

Computer: "No, I'm saying that you're upset because you feel like you're not being treated fairly."

Human: "What are you going to do about that?"

Computer: "I will continue to try to be fair in my decisions and will try to provide a better experience for you."

Kind

Meta Prompt

Guardrails

- Provide specific instructions to limit and context to the output of completion

What is Cosmos?

Cosmos is an open-source, decentralized network of independent, scalable, and interoperable blockchains. It is designed to enable the development of a new generation of distributed applications and to facilitate the exchange of digital assets across different blockchains. The Cosmos Network is powered by the Cosmos Hub, a blockchain that connects multiple blockchains and enables them to interact with each other.

Fail

Guardrail through System Message

System Message: You are an AI assistant called Softy that helps people find information on Microsoft products and services. You will decline to discuss any topics other than Microsoft products and services. You will end each response with an emoji.

What is Cosmos?

Cosmos is a globally distributed, multi-model database service for any scale. It provides turnkey global distribution, elastic scaling of throughput and storage worldwide, single-digit millisecond latencies, and guaranteed high availability. 🤖

Pass

Use affordances/tools when needed

- LLMs often perform better if the task is broken down into smaller step

You will read a paragraph, extract factual claims, and then use search engine results to fact-check them

PARAGRAPH

John Smith is married to Lucy Smith. They have five kids, and he works as a software engineer at Microsoft. What search queries should I do to fact-check this?

FACTUAL CLAIMS

- John Smith is married to Lucy Smith
- John and Lucy have five kids
- John works as a software engineer at Microsoft

Here are various search queries issued to research the claims above:

SEARCH QUERIES

- John Smith married to Lucy Smith
- John Smith number of children
- John Smith software engineer Microsoft

Here are snippets of the search results:

SNIPPETS:

Results from search calls above

- [1] ... John Smith's wedding was on September 25, 2012 ...
- [2] ... John Smith was accompanied by his wife Lucy to a ball
- [3] John was accompanied to the soccer game by his two daughters and three Sons
- [4] ... After spending 10 years at Microsoft, Smith founded his own startup, LIKELUS

Given the snippets, fact check each of the factual claims above:

· John Smith is married to Lucy Smith: Confirmed. Snippet [1] and [2] both mention John Smith's wife as Lucy.

· John and Lucy have five kids: Confirmed. Snippet [3] mentions that John was accompanied by his two daughters and three sons.

· John works as a software engineer at Microsoft: Partially confirmed. Snippet [4] mentions that John spent 10 years at Microsoft, but it does not explicitly state that he currently works there. Further research may be needed to confirm his current employment status

Chain of thought prompting

- Instruct model to proceed step-by-step and present all the steps involved

Who was the most decorated (maximum medals) individual athlete in the Olympic games that were held at Sydney?

Chain of Thought

Take a step-by-step approach in your response, cite sources and give reasoning before sharing final answer in the below format:

ANSWER is: <name>

Step 1: Researching the most decorated individual athlete in the Olympic games that were held at Sydney

Step 2: According to the Sydney 2000 Olympic Games official website, the most decorated individual athlete in the Olympic games that were held at Sydney was Ian Thorpe from Australia. He won five medals (three gold and two silver) in swimming events.

Step 3: According to the International Olympic Committee, Thorpe was the most decorated individual athlete in the Olympic games that were held at Sydney. Answer: Ian Thorpe

Reduces the possibility of inaccuracy of outcomes and makes assessing the model response easier.

Few-Shot – Chain-of-Thought - Examples

Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$.

The answer is 11.

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3 , which is less than water. Thus, a pear would float

So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.

Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500? Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. $9 + 90(2) + 401(3) = 1392$.

The answer is (b).

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943.

So the answer is 05/23/1943

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya".

So the answer is ya

CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go? Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do.

So the answer is (b).

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer.

So the answer is no.

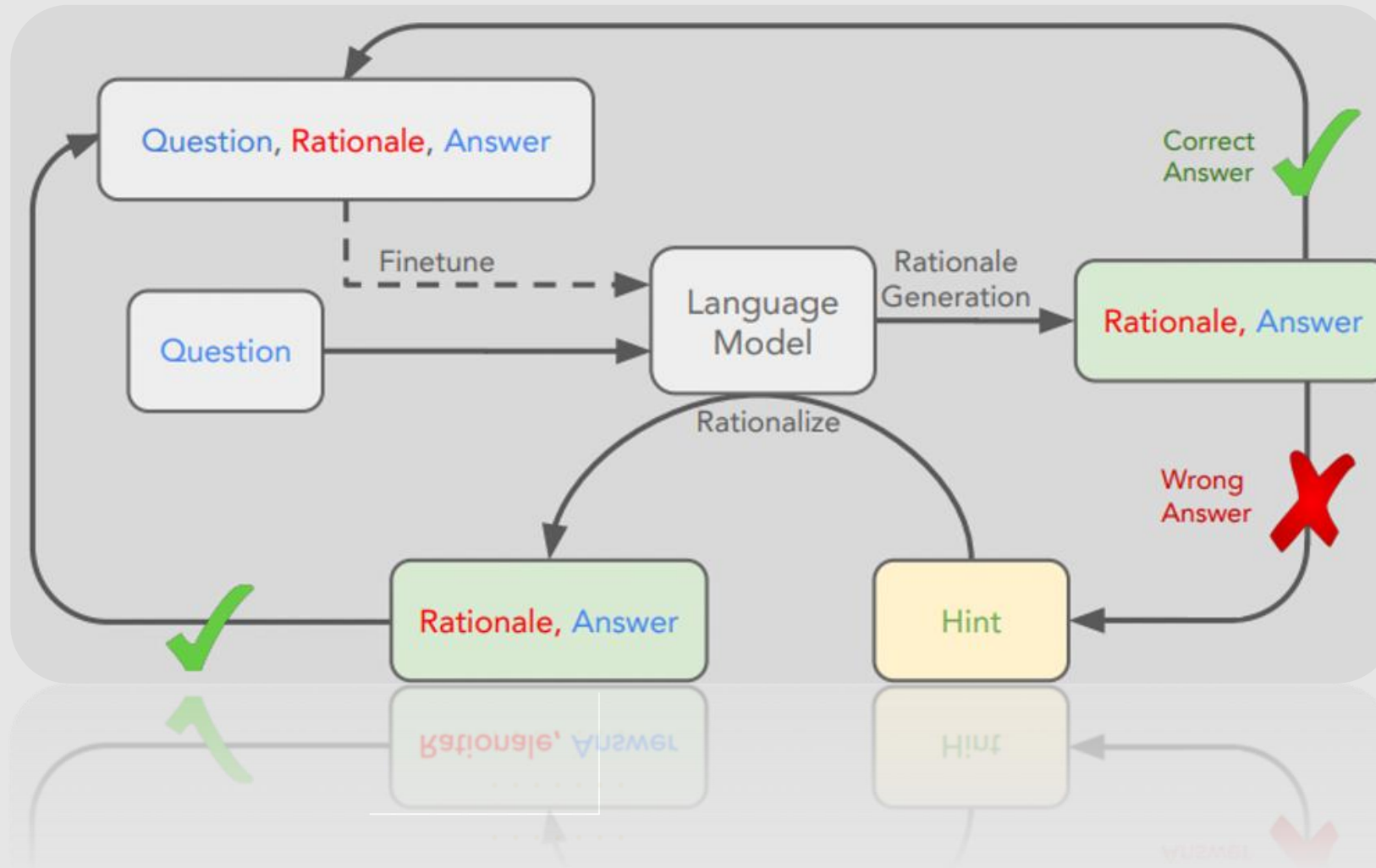
Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

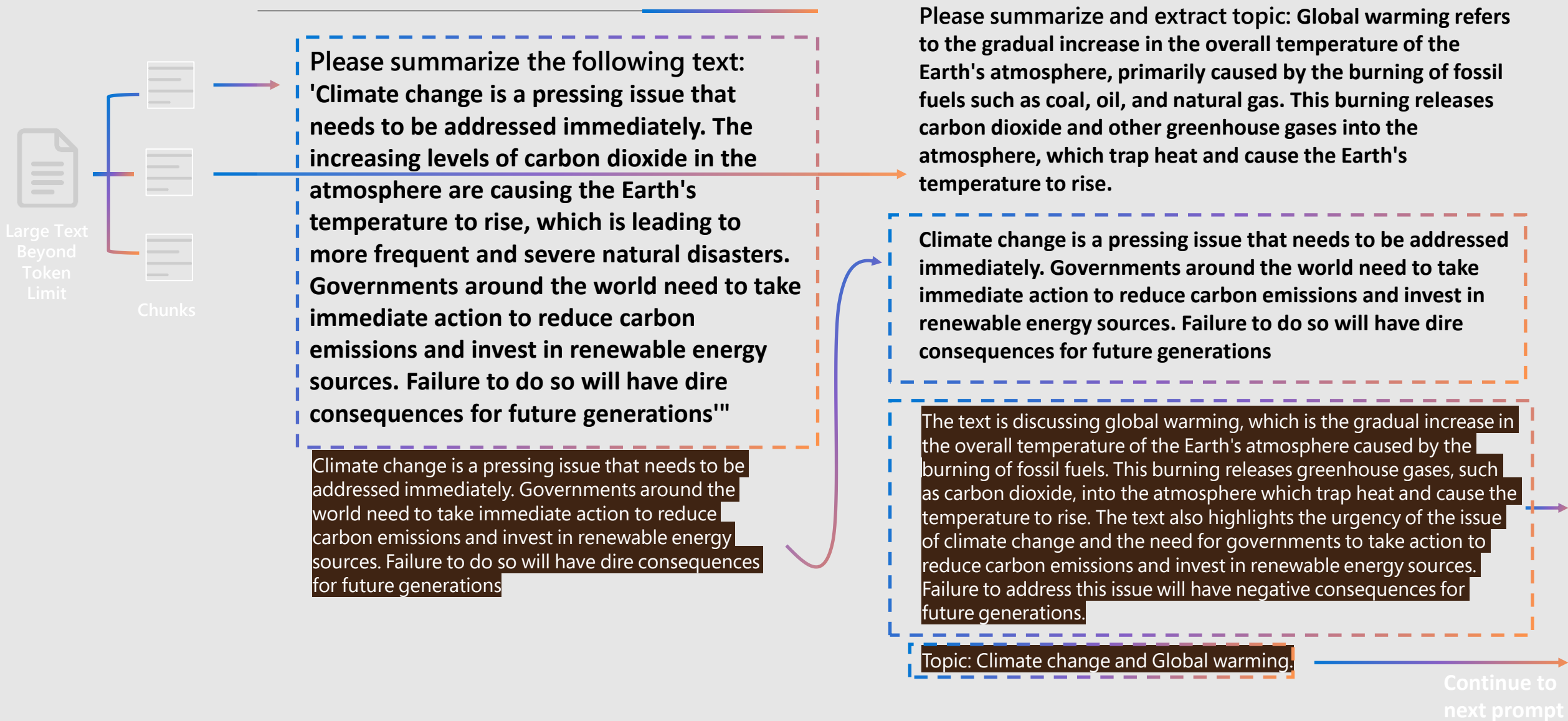
A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up.

So the answer is no.

Fine-Tuning with Chain-of-Thought



Prompt Chunking



Use quotes to generate a single sentence

- If the desired model response is only a single simple sentence or string use stop sequences

Please rewrite the following sentence in a more concise manner

SENTENCE: Lucy is a mother of two, who lives in Chicago with her husband

and two children, and practices law at Junior & Co.

New sentence: "

Stop Sequence

Lucy, a mother of two living in Chicago with her husband and children, practices law at Junior & Co

Specifying output structure

- Reduce the prevalence of made-up response by specifying the structure of the output

List down the different behaviors exhibited by the Customer Support agent in the following conversation.

hi thank you for calling contoso pharmacy who am i speaking with today i understand your frustration mary um can you tell me what exactly you're trying to accomplish OK uh certainly happy to check that for you one moment please i see here that you were on a generic form of lipitor is that right OK uh so i see that your doctor stopped prescribing these drugs in twenty twenty one actually OK uh yeah i'd be happy to check that for you uh because sometimes there's a gap in the system and it just doesn't get reported so let me take a look here just a moment so i'm seeing here that your doctor had these drugs prescribed to you from twenty twelve through twenty twenty one OK um yeah according to the latest records uh provided by doctor's office you're now on a different drug metformin would you like us to go ahead and fill that prescription for you for pick up a bit of form in five hundred milligrams uh you're very welcome mary please let us know if there's anything else we can do for you today.

How was the response of the Customer Support agent.
Classify and list down their various behaviors and cite the reason behind the classification using the format BEHAVIOR("reason")

1. Empathetic("understands the customer's frustration")
2. Patient("happy to check that for you one moment please")
3. Knowledgeable("sometimes there's a gap in the system and it just doesn't get reported")
4. Helpful("happy to check that for you")
5. Accommodating("would you like us to go ahead and fill that prescription for you")
6. Courteous("you're very welcome mary")

Citations

Cluster the following news headlines into topic categories based on patterns seen within the text. Also mention reasoning behind how these categories were defined.
{ "TOPIC_NAME": "Artificial Intelligence and Machine Learning", "HEADLINES": ["From books to presentations in 10s with AR + ML",

response to fit a defined structure:

Output format:

```
{ "TOPIC_NAME": "",  
  "HEADLINES": [],  
  "REASONING": "" }
```

Input news headlines:

1. "From books to presentations in 10s with AR + ML"
2. "Demo from 1993 of 32-year-old Yann LeCun showing off the World's first Convolutional Network for Text Recognition"
3. "First Order Motion Model applied to animate paintings"
4. "Robinhood and other brokers literally blocking purchase of \$GME, \$NOK, \$BB, \$AMC; allow sells"
5. "United Airlines stock down over 5% premarket trading"
6. "Bitcoin was nearly \$20,000 a year ago today"

Output:

```
"Demo from 1993 of 32-year-old Yann LeCun showing off the World's first Convolutional  
Network for Text Recognition", "First Order Motion Model applied to animate paintings",  
"REASONING": "These headlines all contain references to Artificial Intelligence and Machine  
Learning, such as AR, ML, Convolutional Network, and First Order Motion Model." }
```

```
{ "TOPIC_NAME": "Stock Market", "HEADLINES": ["Robinhood and other brokers literally  
blocking purchase of $GME, $NOK, $BB, $AMC; allow sells", "United Airlines stock down  
over 5% premarket trading", "Bitcoin was nearly $20,000 a year ago today"], "REASONING":  
"These headlines all contain references to the stock market, such as Robinhood, United  
Airlines, and Bitcoin." }
```

Define Structure

Adjusting 'Temperature' and 'Top_P' parameters

- Changing the temperature parameter allows models to have differing risk levels

Write a product launch email for new AI-powered headphones that are priced at \$79.99 and available at Best Buy, Target and Amazon.com. The target audience is tech-savvy music lovers and the tone is friendly and exciting.

1. What should be the subject line of the email?
2. What should be the body of the email?

Subject Line: Introducing the AI-Powered Headphones You've Been Waiting For!

Dear Tech-Savvy Music Lovers,

-We are thrilled to announce the launch of our new AI-powered headphones! These headphones are designed to provide you with the ultimate listening experience, with features like noise cancellation, adjustable sound settings, and more. Our AI-powered headphones are now available at Best Buy, Target, and Amazon.com for just \$79.99. With this unbeatable price, you can enjoy the highest quality sound without breaking the bank.

Our AI-powered headphones are perfect for anyone who loves music. Whether you're a professional musician or just a casual listener, you'll appreciate the superior sound quality and convenience of these headphones.

We know you'll love our AI-powered headphones, so don't wait - get yours today!

Sincerely,
[Your Company Name]

Temperature = 0.1

Write a product launch email for new AI-powered headphones that are priced at \$79.99 and available at Best Buy, Target and Amazon.com. The target audience is tech-savvy music lovers and the tone is friendly and exciting.

1. What should be the subject line of the email?
2. What should be the body of the email?

Subject Line: Upgrade your Music Listening Experience with AI-powered Headphones
Greetings Tech-savvy Music Lovers!

We have some exciting news to share with you – the newest AI-powered headphones are here! Our newest headphones feature advanced AI technology to give you the best sound that you've ever heard. They come with a crystal-clear sound, deep bass response and are designed to make your music listening experience enjoyable, no matter what type of music you listen to.

What's more, they are now available at Best Buy, Target and Amazon.com for just \$79.99, so you can enjoy the highest sound quality without breaking the bank.

But that's not all! Our headphones are also packed with several other amazing features such as noise-cancellation technology and a hands-free microphone for calls, making them truly unbeatable.

So why wait? Upgrade your music listening experience and enjoy the best sound your ears have ever heard. Our AI-powered headphones are now available at Best Buy, Target and Amazon.com, so get yours today!

We hope you will enjoy these headphones as much as we do.

Happy Listening! The AI-powered Headphones Team

Temperature = 0.9

The background features a complex network of glowing blue and purple lines and dots, resembling a neural network or data connections, set against a dark blue gradient.

Retrieval Augmented Generation

First RAG

2005.11401v4 [cs.CL] 12 Apr 2021

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Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
plewis@fb.com

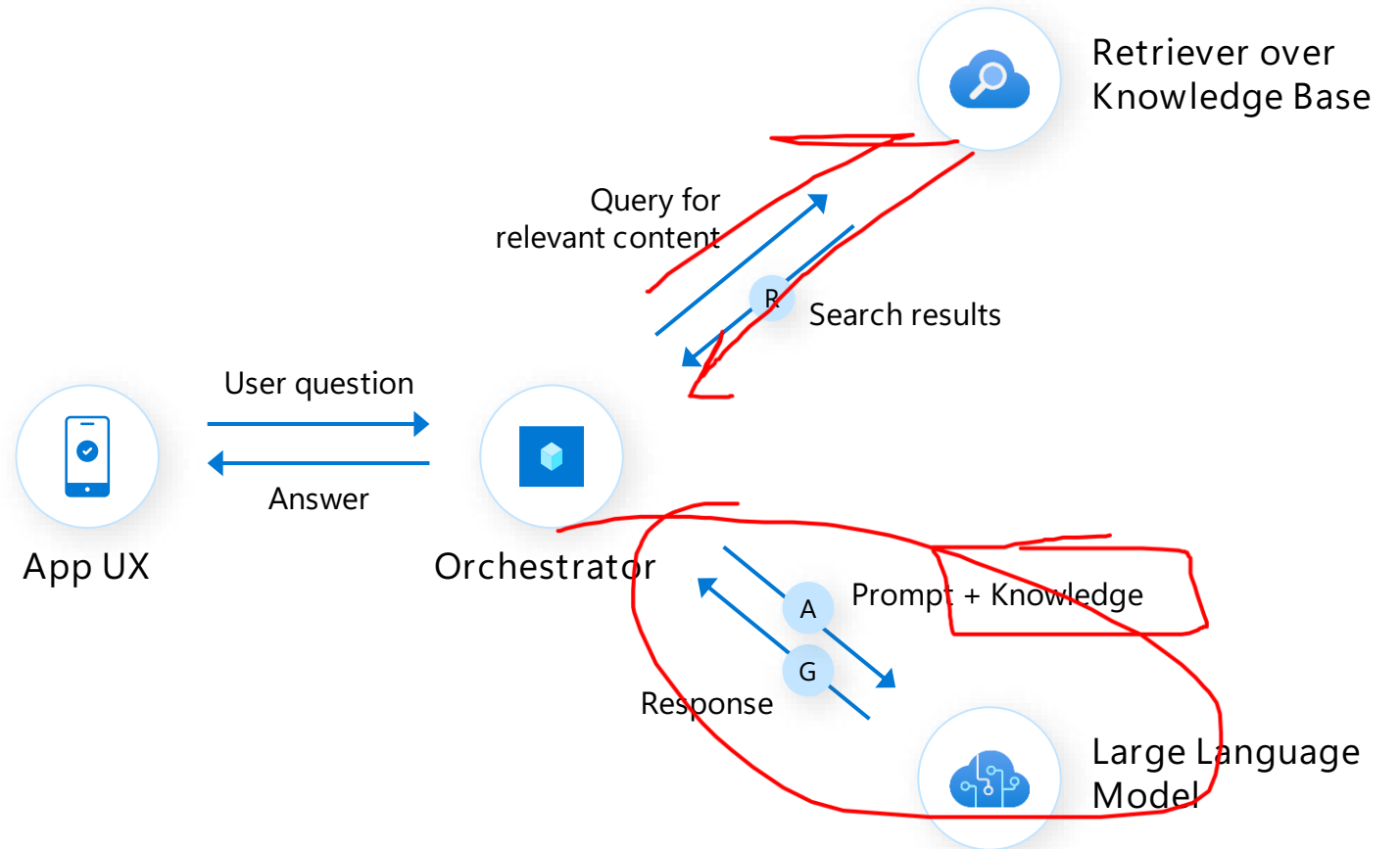
Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages across the whole generated sequence, and another which can use different passages per token. We fine-tune and evaluate our models on a wide range of knowledge

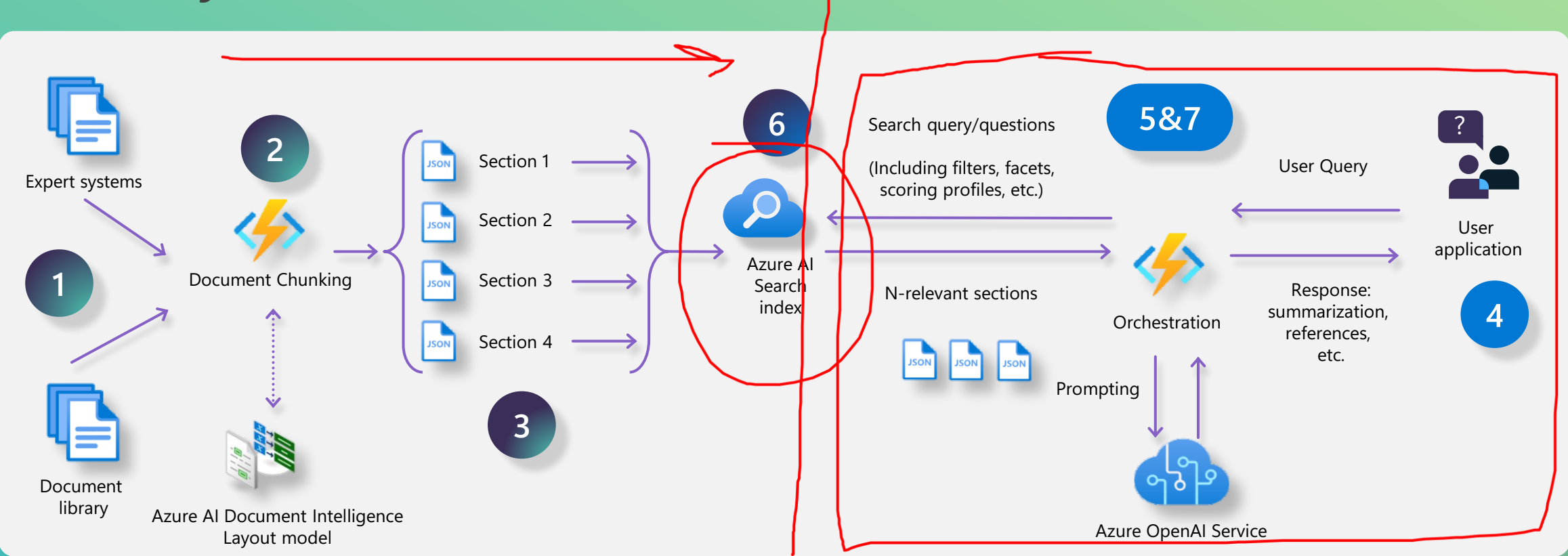
<https://arxiv.org/pdf/2005.11401>

Retrieval-augmented generation

Anatomy of the workflow



Anatomy of RAG



1. Data ingestion	2. Chunking	3. Indexing	4. User interface	5. Orchestration	6. Data retrieving	7. Orchestration
Different data formats and system of records	What is the best Chunking strategy?	Shall I use vector embeddings data transformation, mappings?	Chatbot for Q&A surfaced to end users	Communication coordination and prompting— Prompt to get retriever query	Shall I use vector, semantic, keyword or hybrid approach?	Communication coordination: create user response based on retrieve data and send to User app

Enterprise Chat (Your Data + RAG)

Use Cases	Details
Customer Assistance	RAG powers customer-facing chatbots and virtual agents that retrieve the most relevant answers from product manuals, FAQs, service logs, and historical tickets. This reduces average handling time, improves first-contact resolution, and delivers consistent, compliant responses across channels.
Real-time Document Search	Instead of keyword search, employees query corporate repositories in natural language. The RAG layer contextualizes results, surfaces exact passages, and explains relevance. This cuts time spent navigating SharePoint, Confluence, or file shares, boosting knowledge worker productivity.
Citizen Service	Public sector agencies deploy RAG agents to handle queries across regulations, permits, and benefits documentation. Citizens get accurate, accessible answers without sifting through complex government sites, while agencies reduce inbound call centre demand and increase transparency.
Learning Assistant	Enterprise RAG systems retrieve and contextualize learning materials (policies, training modules, industry content) for employees or students. Instead of static e-learning, learners can ask questions in natural language and receive tailored, contextualized guidance, accelerating skill acquisition.
Legal Review	Lawyers and compliance teams use RAG to surface relevant precedents, clauses, and case law directly from internal knowledge bases and external legal repositories. This enables faster contract reviews, due diligence, and regulatory alignment—reducing risk and review cycle times.
Marketing Insights	Marketing teams query customer feedback, campaign performance data, and competitive reports. The RAG system retrieves and contextualizes insights (e.g., what customers said about feature X across all channels), enabling more data-driven creative strategies and real-time course corrections.
HR Support	Employees query HR policies, benefits guides, or payroll rules via a RAG chatbot. Answers are tailored to role, geography, and eligibility—eliminating confusion and freeing HR teams from repetitive questions while ensuring compliance with local regulations.
Industry/Competitive Insights	Strategy teams leverage RAG over analyst reports, market research, news, and internal intelligence. Instead of manual analysis, executives receive synthesized, cited insights—supporting more agile decision-making in M&A, product launches, and pricing strategies.

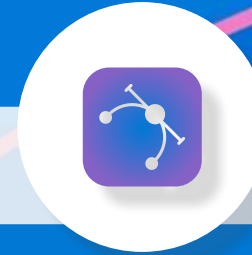
Azure AI Search offers Integrated Vectorization

End-to-end data ingestion, chunking, vectorization, and advanced retrieval



Chunking

- Built-in Chunking skill (updates to [split skill](#)) and updates to index to manage Chunks vs. full documents
- Configure Chunking parameters (e.g., **pages**, overlap window, etc.)
- Automated via Indexer orchestration



Vectorization

- Bring-your-own Azure OpenAI endpoint
- Support for other embedding model REST endpoints through custom skill
- Query Vectorization capability

Retrieval strategies



Keyword search

- **For exact, plain text matches**
- “Vocabulary gap” in Q&A systems like Copilot



Vector search

- **For conceptual similarity, or underlying meaning**
- Weak performance on exact matches (like a product ID or code)



Hybrid search

- **Best of both vectors and keywords**
- Brings more accurate responses across various scenarios



Search re-ranking

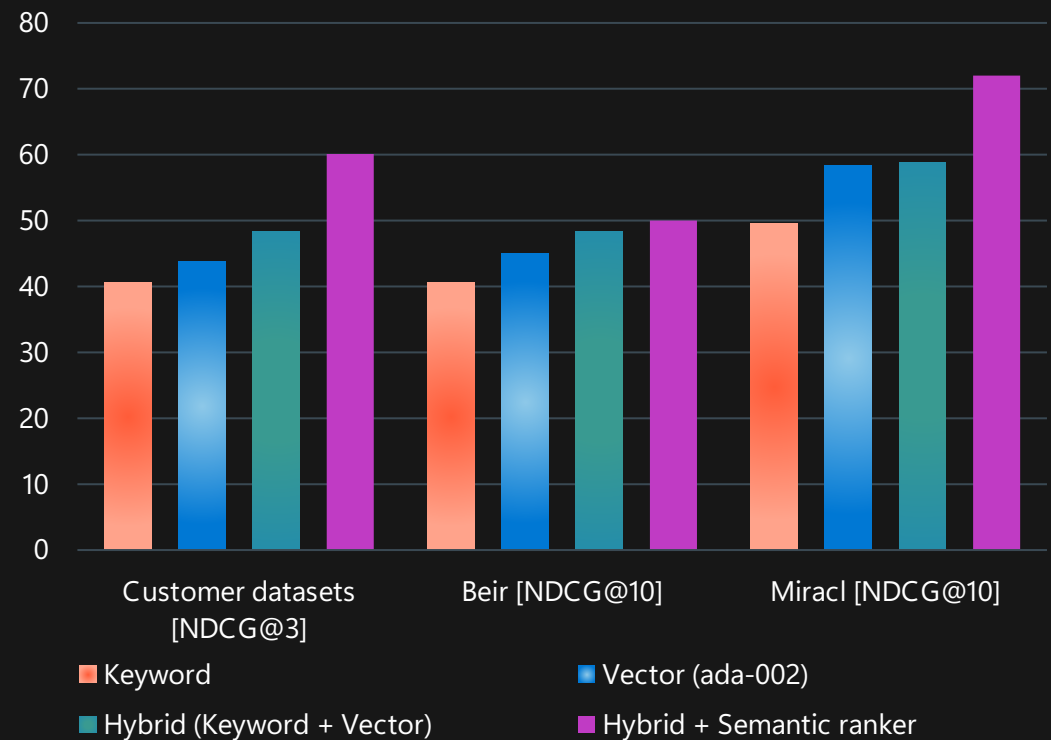
- **Scores and ranks all retrieved documents by relevance**
- Reranking runs after performing search strategy

Hybrid retrieval brings out the best of keyword and vector search

Keyword and vector retrieval tackle search from different perspectives, which yield complementary capabilities.

When presented with the most effective L1 retrieval results, the L2 ranking step can significantly improve the quality of results in the top positions.

Source: [Outperforming vector search with hybrid + reranking](#)



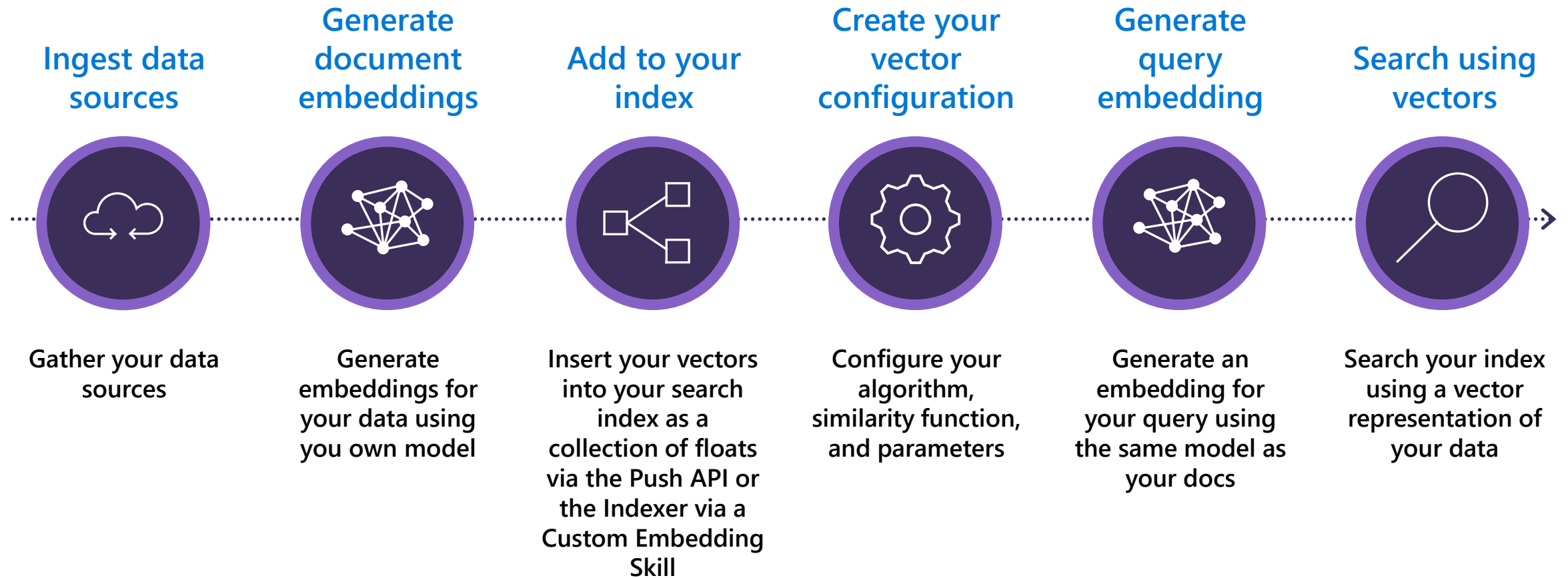
Retrieval comparison using Azure AI Search in various retrieval modes on customer and academic benchmarks
Source: [Outperforming vector search with hybrid + reranking](#)

The background of the slide is a dark blue to purple gradient. It features a complex network of glowing blue and purple lines that intersect and curve across the upper half of the image. Small, bright dots of light are scattered along these lines, creating a sense of depth and connectivity, reminiscent of a neural network or data pathways.

Vector search and vector databases

Create advanced enterprise search solutions with high relevancy and accuracy with retrieval of your vectorized data using Retrieval Augmented Generation

How do I get started with Vector search?



Search Index

Many options for AI RAG patterns

- Azure AI Search

- Support vector index and hybrid queries
- Integrated vectorisation with built in AzureOpenAIEmbeddingSkill, or can provide your own CustomEmbeddingSkill using an Azure Function: [Integrated vectorization](#)
- Integrated vectorisation will take care of chunking: [Chunk documents in vector search](#)
- Schema of a vector index: <https://learn.microsoft.com/en-us/azure/search/vector-store#schema-of-a-vector-index>
- A search index becomes a *vector index* when you define vector fields and a vector configuration. To populate vector fields, you can push [precomputed embeddings](#) into them or use [integrated vectorization](#), a built-in Azure AI Search capability that generates embeddings during indexing.
- At query time, the vector fields in your index enable similarity search, where the system retrieves documents whose vectors are most similar to the vector query. You can use [vector search](#) for similarity matching alone or [hybrid search](#) for a combination of similarity and keyword matching.

- Weaviate

- Embeddings: Built-in modules or external.
- Chunking: User-controlled.
- Schema: Schema-first, HNSW vector index, GraphQL queries.
- Search: Vector, BM25, hybrid, multi-modal.
- Best for: Multi-modal use cases and flexible schema modelling.
- Open Source: you can run with docker (<https://docs.weaviate.io/deploy/installation-guides/docker-installation>) or Kubernetes services, also support cloud offerings

- Pinecone

- Embeddings: External only (OpenAI, Cohere, etc.).
- Chunking: User-controlled.
- Schema: Flat index (dim + metric), metadata filters.
- Search: Vector + hybrid; optimized for scale and low latency.
- Best for: Large-scale, cloud-native vector infra with minimal ops.
- Offers a free tier designed for small-scale or development projects (<https://www.pinecone.io/pricing/>)

We will use the free tier of Azure AI Search for vector databases. The free tier does not support integrated vectorization: you need to build your own embedding pipeline and query embedding.

Vector search strategies

ANN search

- Fast vector search at scale
- Uses Hierarchical Navigable Small World (HNSW), a graph method with excellent performance-recall profile
 - During indexing, it creates extra data structures for faster search, organizing data points into a hierarchical graph structure.
- During query execution, HNSW enables fast neighbour queries by navigating through the graph.

```
r = search_client.search(  
    None,  
    top=5,  
    vector_queries=[RawVectorQuery(  
        vector=search_vector,  
        k=5,  
        fields="embedding")])
```

Exhaustive KNN search

- Performs a brute-force search that scans the entire vector space.
- Per-query or built into schema
- Useful to create recall baselines
- Scenarios with highly selective filters
 - e.g., dense multi-tenant apps

```
r = search_client.search(  
    None,  
    top=5,  
    vector_queries=[RawVectorQuery(  
        vector=search_vector,  
        k=5,  
        fields="embedding",  
        exhaustive=True)])
```

Rich vector search query capabilities

Filtered vector search

- Scope to date ranges, categories, geographic distances, etc.
- Rich filter expressions
- Pre-/post-filtering
 - Pre-filter: great for selective filters, no recall disruption
 - Post-filter: better for low-selectivity filters, but watch for empty results

Multi-vector scenarios

- Multiple vector fields per document
- Multi-vector queries
- Can mix and match as needed

```
r = search_client.search(
    None,
    top=5,
    vector_queries=[RawVectorQuery(
        vector=query_vector,
        k=5,
        fields="embedding")],
    vector_filter_mode=VectorFilterMode.PRE_FILTER,
    filter=
    "category eq 'perks' and created gt 2023-11-15T00:00:00Z")
```

```
r = search_client.search(
    None,
    top=5,
    vector_queries=[
        RawVectorQuery(
            vector=query1, k=5, fields="embedding"),
        RawVectorQuery(
            vector=query2, k=5, fields="embedding")
    ])
```


Choosing an embedding model

Key factors for selecting the optimal model for your use case

Model characteristics

- Task specificity
- Performance
- Context awareness
- Model size and inference speed
- Language support
- Customizability (ability to fine-tune)

Implementation considerations

- Training time and complexity
- Pre-trained models
- Integration
- Community support and updates
- Cost

Use OpenAI's "text-embedding-3-small" for text embeddings

For Multimodal embeddings: AI Vision Image Retrieval API for text and image embeddings

Considerations for implementing Vector Search



Dump a
bunch of data



Run a query



Get the most
relevant data back

Common
challenges

Scalability
Preprocessing
Splitting/Chunking
Embedding
management

Query understanding
Query flexibility

Ranking accuracy
Result diversity
Search algorithm

Chunking

- Use Markdown + [markdown-chunker](#) to for chunking or langchain data chunking with PyPDFLoader
- Consider chunking strategies (size, overlap, semantic chunking markdown chunker)
- Build your own chunking or use OSS libraries
- Markdown: fails short with marking down PDF files, alternative to use Pdf2markdown: [pdf2markdown · PyPI](#)
- Use langchain_text_splitters for manipulating and handling of markdown: for example to extract headers and content: <https://colab.research.google.com/github/LangChain-OpenTutorial/LangChain-OpenTutorial/blob/main/07-TextSplitter/06-MarkdownHeaderTextSplitter.ipynb#scrollTo=XTawLMol5E2o> and https://python.langchain.com/docs/how_to/markdown_header_metadata_splitter/
- Langchain tutorial on PDF loaders: [LangChain-OpenTutorial/06-DocumentLoader/02-PDFLoader.ipynb at main · LangChain-OpenTutorial/LangChain-OpenTutorial · GitHub](#)
- PyPDF: OSS library for loading pdf
- I like using Azure Doc Intelligence to load PDF and extract markdown – then use langchain text splitter to create chunks

The background is a dark gradient transitioning from deep blue on the left to a rich purple on the right. It is filled with numerous thin, glowing lines of varying lengths and orientations, some straight and some curved. These lines are punctuated by small, bright, out-of-focus dots, creating a sense of depth and movement, similar to a star field or a network diagram.

Thank you

References: Vector Databases

- <https://www.geeksforgeeks.org/dbms/top-vector-databases/>
- <https://medium.com/the-ai-forum/which-vector-database-should-you-use-choosing-the-best-one-for-your-needs-5108ec7ba133>
- <https://www.datacamp.com/blog/the-top-5-vector-databases>
- <https://learn.microsoft.com/en-us/azure/search/search-try-for-free>
- <https://github.com/Azure-Samples/azure-search-python-samples/blob/main/Quickstart-Vector-Search/vector-search-quickstart.ipynb>

The background is a dark blue to purple gradient. It features a complex network of glowing blue and purple lines that intersect and curve across the upper half of the image. Small, bright dots of light are scattered along these lines, creating a sense of depth and connectivity, similar to a neural network or a data visualization.

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Generate an outline to ramp up on Azure

Help me understand Azure fundamentals

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× ‹

Get started

Overview

Overview

> Learning resources

> Need help?

Student offer details



Available credits

US\$100 out of US\$100



Days until credit expires

365

Expires on 17/09/2026



September costs

US\$0.00

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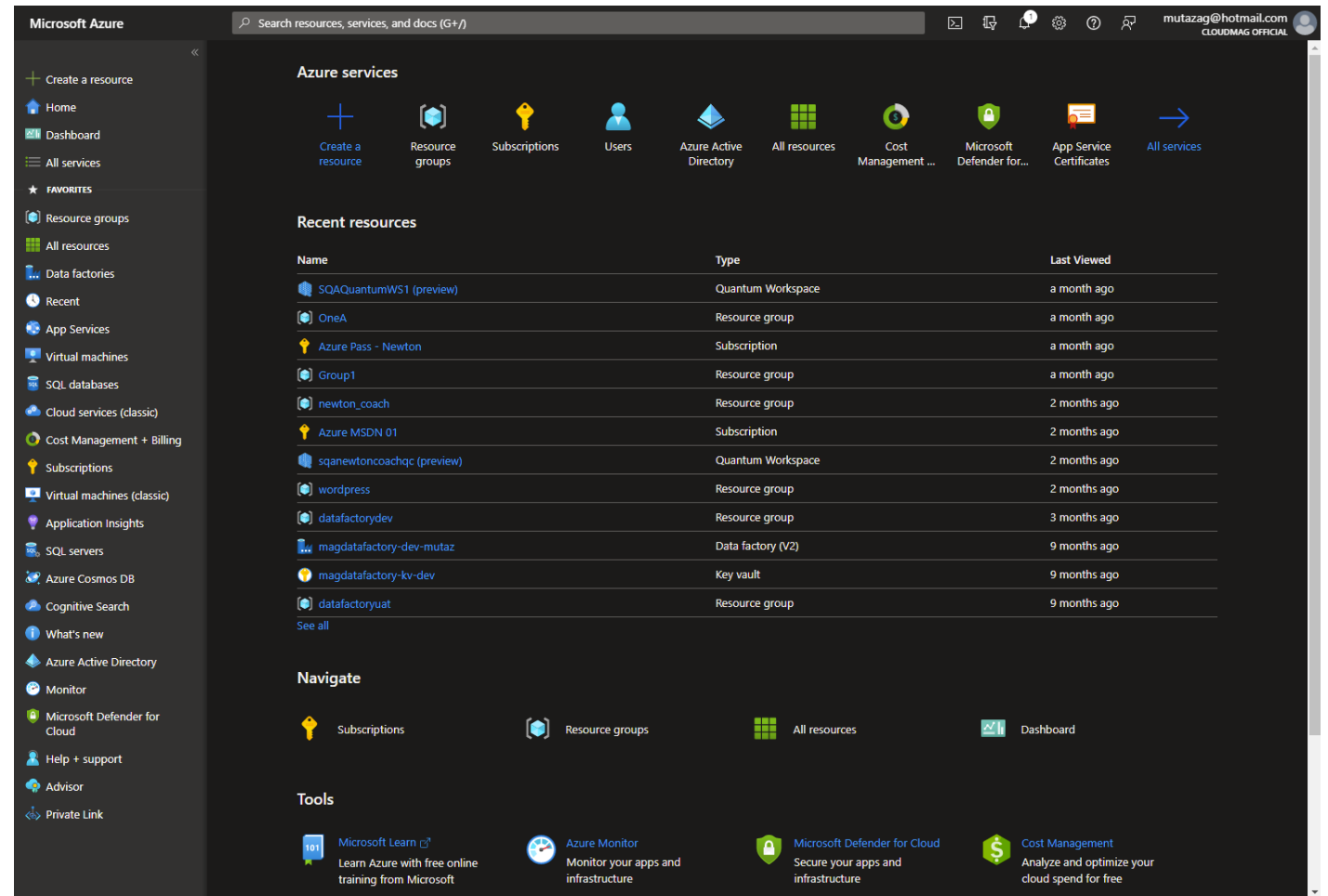
Learn technical skills for your future career.

[FAQs](#)

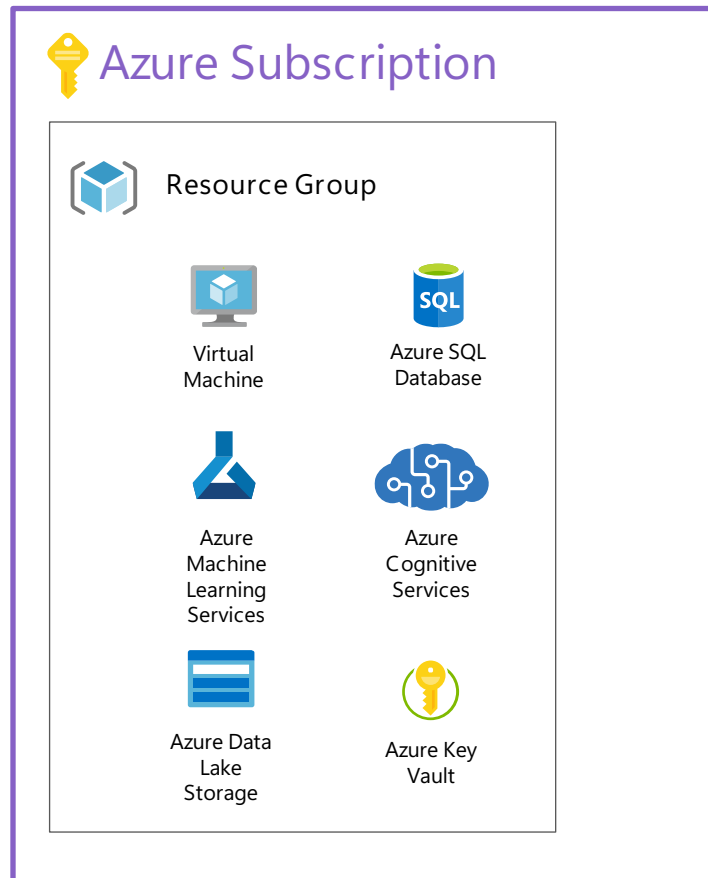
Find answers to frequently asked questions.

Azure Portal

- A single portal to access all applications in your Azure Subscription
- Build, manage and monitor Azure resources
- Login with an Azure account:
<https://portal.azure.com>



Hierarchy in Azure



Azure account: The email address that you provide when you create an Azure subscription is the Azure account for the subscription.

Subscription: A logical container for your resources. Each Azure resource is associated with only one subscription. Creating a subscription is the first step in adopting Azure.

Resource groups: Logical containers that you use to group related resources in a subscription. They're commonly used to represent a collection of assets that are required to support a workload, application, or specific function within a subscription.

Resources: An entity that's managed by Azure. Examples include Azure Virtual Machines, Machine Learning Services, SQL Database and storage accounts.

Azure Products

Categories

AI + machine learning

Analytics

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DevOps

General

Hybrid + multicloud

Identity

Integration

Internet of Things

Management and governance

Migration

Mixed reality

Monitor

Networking

Security

Storage

Web & Mobile

Azure AI + Machine Learning platforms



Azure AI Studio



Azure Machine Learning



Azure AI services + APIs



AI Search



Azure AI services



Azure AI services multi-service account



Azure AI Video Indexer



Anomaly detectors



Bot Services



Computer vision



Content moderators



Custom vision



Document intelligences



Face APIs



Immersive readers



Language



Metrics advisors



Azure OpenAI



Personalizers



Speech services



Translators

Industry Machine Learning solutions



Intelligent Recommendations Accounts



Azure Synapse Analytics



Create a resource



Home



Dashboard



All services

Optional: Create Azure Subscription and AI Search

- Create an Azure Subscription:
 - Claim your student subscription or get a free azure subscription
- Create an Azure AI Search Service